

OSSP PRACTICAL 11

Practical 11 — Theory

Advanced Synchronization and Concurrency Hazards

Synchronization is used in operating systems to control the execution of multiple threads or processes that access shared resources. Without proper synchronization, simultaneous access can produce incorrect or unpredictable results.

A **critical section** is the part of a program where a shared resource is accessed or modified. Only one thread should execute the critical section at a time when the shared resource requires exclusive access.

A **mutex (mutual exclusion)** is a synchronization mechanism that allows only one thread to access a critical section at a time. In our program, the mutex protects the shared counter variable.

Important Functions

- `pthread_create()` — creates a new thread.
- `pthread_join()` — waits for a thread to finish.
- `pthread_mutex_init()` — initializes a mutex.
- `pthread_mutex_lock()` — locks the mutex before entering the critical section.
- `pthread_mutex_unlock()` — releases the mutex after leaving the critical section.
- `pthread_mutex_destroy()` — destroys the mutex after use.

Concurrency Hazards

When multiple threads access shared data without proper synchronization, problems such as **race conditions** can occur. A race condition happens when the final result depends on the timing or order of thread execution.

Using a mutex prevents simultaneous modification of the shared variable and ensures consistent results.

Valgrind

Valgrind is used to detect memory-related problems. In this practical, it was used to check for memory leaks and errors.

Our result was:

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Final counter = 200000

All heap blocks were freed -- no leaks are possible

ERROR SUMMARY: 0 errors from 0 contexts

Conclusion

Synchronization mechanisms such as mutexes provide safe access to shared resources in multithreaded programs. Proper synchronization prevents race conditions and helps ensure correct and predictable program execution.

```

alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical11
alurusaranya@LAPTOP-0PA61QEN:~$ cd ~
alurusaranya@LAPTOP-0PA61QEN:~$ mkdir practical11
mkdir: cannot create directory 'practical11': File exists
alurusaranya@LAPTOP-0PA61QEN:~$ pwd
/home/alurusaranya
alurusaranya@LAPTOP-0PA61QEN:~$ cd practical11
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ pwd
/home/alurusaranya/practical11
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ nano sync.c
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ls -l
total 4
-rw-rw-r-- 1 alurusaranya alurusaranya 599 Sep 28 05:20 sync.c
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ gcc sync.c -o sync -pthread
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ./sync
Final counter = 200000
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ./sync
Final counter = 200000
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ./sync
Final counter = 200000
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ./sync
Final counter = 200000
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ valgrind --version
valgrind-3.22.0
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ valgrind ./sync
==849== Memcheck, a memory error detector
==849== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==849== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==849== Command: ./sync
==849==
Final counter = 200000
==849==
==849== HEAP SUMMARY:
==849==   in use at exit: 0 bytes in 0 blocks
==849==   total heap usage: 3 allocs, 3 frees, 1,568 bytes allocated
==849==
==849== All heap blocks were freed -- no leaks are possible

==849==
==849== All heap blocks were freed -- no leaks are possible
==849==
==849== For lists of detected and suppressed errors, rerun with: -s
==849== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ valgrind --leak-check=full ./sync
==852== Memcheck, a memory error detector
==852== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==852== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==852== Command: ./sync
==852==
Final counter = 200000
==852==
==852== HEAP SUMMARY:
==852==   in use at exit: 0 bytes in 0 blocks
==852==   total heap usage: 3 allocs, 3 frees, 1,568 bytes allocated
==852==
==852== All heap blocks were freed -- no leaks are possible
==852==
==852== For lists of detected and suppressed errors, rerun with: -s
==852== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ps -elf | head
UID          PID    PPID    LWP   C NLWP STIME TTY          TIME CMD
root           1         0      1  0   1 04:49 ?        00:00:01 /sbin/init
root           2         1      2  0   2 04:49 hvc0    00:00:00 /init
root           2         1      8  0   2 04:49 hvc0    00:00:00 /init
root           6         2      6  0   2 04:49 hvc0    00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --
log-truncate
root           6         2      7  0   2 04:49 hvc0    00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --
log-truncate
root          52         1     52  0   1 04:49 ?        00:00:00 /usr/lib/systemd/systemd-journald
root         100         1    100  0   1 04:49 ?        00:00:00 /usr/lib/systemd/systemd-udev
systemd+     111         1    111  0   1 04:49 ?        00:00:00 /usr/lib/systemd/systemd-resolved
systemd+     129         1    129  0   2 04:49 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ps -elf | grep sync
systemd+     129         1    129  0   2 04:49 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd
systemd+     129         1    158  0   2 04:49 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd

```

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```
systemd+ 129 1 129 0 2 04:49 ? 00:00:00 /usr/lib/systemd/systemd-timesyncd
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ps -elf | grep sync
systemd+ 129 1 129 0 2 04:49 ? 00:00:00 /usr/lib/systemd/systemd-timesyncd
systemd+ 129 1 158 0 2 04:49 ? 00:00:00 /usr/lib/systemd/systemd-timesyncd
alurusa+ 860 584 860 0 1 05:22 pts/0 00:00:00 grep --color=auto sync
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ rm sync
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ ls -l
total 4
-rw-rw-r-- 1 alurusaranya alurusaranya 599 Sep 28 05:20 sync.c
alurusaranya@LAPTOP-0PA61QEN:~/practical11$ |
```

```
GNU nano 7.2 sync.c
#include <stdio.h>
#include <pthread.h>

int counter = 0;
pthread_mutex_t lock;

void *increment(void *arg)
{
    for (int i = 0; i < 100000; i++)
    {
        pthread_mutex_lock(&lock);

        counter++;

        pthread_mutex_unlock(&lock);
    }

    return NULL;
}

int main()
{
    pthread_t t1, t2;

    pthread_mutex_init(&lock, NULL);

    pthread_create(&t1, NULL, increment, NULL);
    pthread_create(&t2, NULL, increment, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Final counter = %d\n", counter);

    pthread_mutex_destroy(&lock);

    return 0;
}
```